Every formula has a formation sequence

The Open Logic Project contributors

openlogic:propositional-logic/syntax-and-semantics:local-snapshot-20261013; CC BY 4.0

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Open Logic Project unit file sources/olp-propositional-logic-syntax-and-semantics/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. All five connective cases are carried explicitly (tagitem enumerate), so the counted proof is complete at the author's level.

Editorial note. A same-label proposition in the first-order-logic-syntax-and-semantics unit is a distinct unit-scoped statement (quantifier cases) and is a separate registered card (L46214); cross-unit same-name labels are chapter-scoped per the fitting-ideal precedent.

Editorial note. The label prop:formed names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Allowed standard prerequisites (editorial).

- Principle of induction on formulas. Same unit, \ollabel{thm:induction} (source line 950, own card). Lines 618-619: closes the five constructed cases into the universal claim.

Proposition 0.1. *Every formula A in* $\mathrm{Frm}(\mathcal{L}_0)$ *has a formation sequence.*

Proof. Suppose A is atomic. Then the sequence $\langle A \rangle$ is a formation sequence for A. Now suppose that B and C have formation sequences $\langle B_0, \ldots, B_n \rangle$ and $\langle C_0, \ldots, C_m \rangle$ respectively.

1. If $A \equiv \neg B$, then $\langle B_0, \ldots, B_n, \neg B_n \rangle$ is a formation sequence for A.
2. If $A \equiv (B \wedge C)$, then $\langle B_0, \ldots, B_n, C_0, \ldots, C_m, (B_n \wedge C_m) \rangle$ is a formation sequence for A.
3. If $A \equiv (B \vee C)$, then $\langle B_0, \ldots, B_n, C_0, \ldots, C_m, (B_n \vee C_m) \rangle$ is a formation sequence for A.
4. If $A \equiv (B \to C)$, then $\langle B_0, \ldots, B_n, C_0, \ldots, C_m, (B_n \to C_m) \rangle$ is a formation sequence for A.
5. If $A \equiv (B \leftrightarrow C)$, then $\langle B_0, \ldots, B_n, C_0, \ldots, C_m, (B_n \leftrightarrow C_m) \rangle$ is a formation sequence for A.

By the principle of induction on formulas, every formula has a formation sequence. □